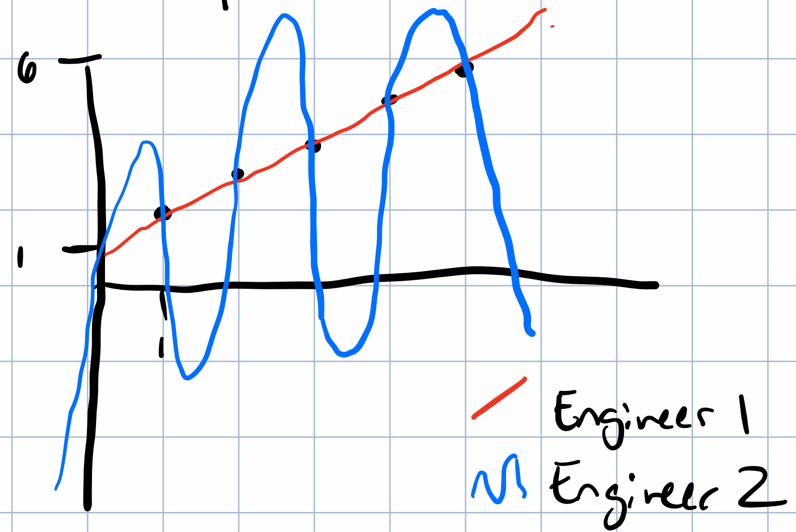
1.1) Suppose you are given 5 points of data:

x	y
1	2
2	3.1
3	3.9
4	5.0
5	5.8



Suppose then we want to predict x=6. Two engineers propose two different models.

Engineer 1 proposes a linear relationship

$y = ax + b$

Engineer 2 proposes a fifth degree polynomial

$y = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x^1 + a_0x^0$

Engineer 2's method passes exactly through each point exactly once.

Which do you trust more?

I trust Engineer 1's model more. 2's model over-fits the data. I'm tempted to solve it just to see the shape it makes, but I know the curve will be all over the place.

Using a fifth degree polynomial for five pieces of data will give an exact solution every time, but that tells us almost nothing.

Hyphotheses need to pay rent. A model that can fit anything tells us nothing. The fifth-degree polynomial would predict the exact values at those points, but give terrible results farther away from those exact points.

The exact points don't matter, since our experimental data contains noise. Engineer 1's model takes that into account. Engineer 2's model pretends the noise <u>is</u> data.

6.1) Suppose a model achieves

- training accuracy 100%
- test accuracy 58%

What does that signal?

Our model has over-fit to the training data. It hasn't learned the right abstractions.

It's not unlike a student who has a test coming up gets a copy of the same test from past year and memorizes the answers, rather than learning the material. If the new test is changed, that student is not going to perform well against one that learned the right techniques to produce the right answers.

6.2) Suppose another model achieves

- training accuracy 91%
- test accuracy 90%

Which model would you deploy? Why?

Model 2 will be a lot more useful. It's test accuracy we ultimately care about, not training.

6.3) Suppose I double the parameters in a neural network, must the test accuracy improve? Why or why not?

No. As a matter of fact the model will have a higher capacity for memorizing the training data, meaning that it risks over-parameterizing the data.

6.4) Suppose you are fitting points with increasingly higher and higher degree polynomials. At what point does adding more degrees stop helping? How would you know?

The exact point will depend on what you are modeling.

For instance a forier transform can model a square wave using just sin and cos. At the corners things get weird but as you continue the result gets closer and closer to the real result, (and in the infinate case becomes exact).

However, in our case, we should be able to hold back some of our training data and use it to test the model, and find the point where performance degrades from over-fitting.

We can also introduce <u>more</u> error in our training protocol, such that each time that data point comes up, it will be at some random point centered on the actual value. That would force the model to not just memorize the training data as there will be a fundimental limit to how accurate it can be. However if we introduce a normal distribution of error, the error will average out over successive tests, assuming the model can identify the underlying structure of the data, and be more resiliant to noise.